1.4 Trigonometric Functions of Any Angle

Name _____ Period ____ Date _____

Pre-Calculus 1.4 (30-minute plan) · Student sheet

I can…

- evaluate trigonometric functions from a point on the terminal side.
- find the sign of a trigonometric function from the quadrant.
- find reference angles and use them to find exact values.

Vocabulary · fill in the missing word

Reference angle — the acute angle θ′ between the _____ side of θ and the x-axis

Quadrantal angle — an angle whose _____ side lies on an axis

Part 1 · Notes and examples (fill in each note, then work the example)

Any angle: for (x, y) on the terminal side, $r = \sqrt{(x^2 + y^2)}$: $\sin\theta = y/r$, $\cos\theta =$ _____, $\tan\theta = y/x$.

Example 1

$(-3, 4)$ is on the terminal side of θ. Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Reference angle θ′: the acute angle between the terminal side and the _____.

Example 2

Find the reference angle.

a. 300°

b. 2.3

c. −135°

To evaluate: find the value for θ′, then give it the _____ of the quadrant.

Example 3

Evaluate.

a. $\cos\frac{4\pi}{3}$

b. $\tan(-210°)$

c. $\csc\frac{11\pi}{4}$

Part 2 · Partner practice

Partner practice

Evaluate sin 210°.